Java Assignment 10 - Marks 20

1 Bank Transaction Processor

Build an Account hierarchy and process transactions given as strings.

(a) Create an abstract class `Account` with fields `accountNo`, `holderName` and `balance`, and an abstract method `withdraw(double amt)`.

(b) Create two subclasses:

- `SavingsAccount`: minimum balance Rs. 1000, and at most 3 withdrawals per session.
- `CurrentAccount`: overdraft allowed up to Rs. 10,000.

(c) Create the following custom exceptions:

- `InsufficientFundsException` (checked), which stores the shortfall amount.
- `WithdrawalLimitExceededException` (checked).
- `InvalidTransactionException` (unchecked), for malformed input.

(d) Read transactions as strings in the format `TYPE:ACCNO:AMOUNT`, for example `DEP:S101:5000`, `WDR:C202:12000` or `TRF:S101:C202:300`.

(e) Parse each one using `split()` and `trim()`. Reject invalid types, missing fields and non-numeric amounts. Wrap any `NumberFormatException` inside `InvalidTransactionException` using **exception chaining**.

(f) Process all transactions, even when some fail. At the end, print a report of how many succeeded and how many failed, with the reason for each failure.

Employee Payroll

(a) Create an abstract class `Employee` with abstract method `calculateSalary()`. Add three subclasses:

- `Manager`: base pay plus bonus.
- `Engineer`: base pay plus (overtime hours × rate).
- `Intern`: fixed stipend.

(b) Input lines look like: `"E102, Ravi Kumar , ENGINEER, 45000, 12"`

(c) Validate each line:

- The ID must match the pattern `E\d{3}`. Use `matches()`.
- The name may contain only letters and spaces, and must be stored in Title Case. Write your own method for this without using any library.
- Salary must be positive. Throw a custom `InvalidEmployeeDataException` that records the **line number** and the **field name**.

(d) Use a `StringBuilder` to produce a formatted payslip table with aligned columns using `String .format()`.

(e) Sort the employees by salary and print the top earner of each type using `instanceof` or `getClass()`.

Note: *The regular expression pattern `E101` is used in Java to validate standard employee identification codes.*